

Inverses

An $n \times n$ matrix A is said to be invertible if there is an $n \times n$ matrix C , such that

$$CA = I \text{ and } AC = I$$

where $I = I_n$, the $n \times n$ identity matrix. In this case C is an inverse of A and is uniquely determined by A . This uniquely determined inverse matrix is denoted A^{-1} , so that

$$A^{-1}A = I \text{ and } AA^{-1} = I$$

A non-invertible matrix is sometimes called a singular matrix. And an invertible matrix is called a non-singular matrix

Example

$$A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}, C = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}$$

$$AC = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -14+15 & -10+10 \\ 21-21 & 15-14 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$CA = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} = \begin{bmatrix} -14+15 & -35+35 \\ 6-6 & 15-14 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Thus } C = A^{-1}$$

Theorem

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, if $ad-bc \neq 0$, then A is invertible

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad-bc = 0$, then A is not invertible.

Inverses

II

Theorem

If A is an invertible $n \times n$ matrix, then for each $\vec{b}$ in $\mathbb{R}^n$ the equation $A\vec{x} = \vec{b}$ has the unique solution

$$\vec{x} = A^{-1}\vec{b}$$

Suppose we have the system

$$3x_1 + 4x_2 = 3$$

$$5x_1 + 6x_2 = 7$$

Let's put this on $A\vec{x} = \vec{b}$ form and multiply by the inverse on each side, starting with finding the inverse.

$$A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad A^{-1} = \frac{1}{18-20} \begin{bmatrix} 6 & -4 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}$$

Inserting into the equation

$$\vec{x} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} -9 + 14 \\ 15/2 - 21/2 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$$

How do we find inverses for larger matrices?

One way is to set up the augmented matrix

$$[A \mid I]$$

and perform row ops until I has replaced A .

$$[A \mid I] \xrightarrow{\text{G-J}} [I \mid A^{-1}] \quad A^{-1} = \begin{bmatrix} -1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \leftarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right]$$

$$R_2 \leftarrow R_2 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right]$$

$$R_1 \leftarrow R_1 - 2R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -2 & 2 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right]$$

$$A^{-1}A = \begin{bmatrix} -1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1+2 & -2-2+4 & -2+2 \\ 1-1 & 2+1-2 & 1-1 \\ -1+1 & -2+2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$